



ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC KINH TẾ - LUẬT

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
UNIVERSITY OF ECONOMICS AND LAW

DATA MINING

Lecture 2: Data Warehouse Technology



Lecture 2 outline

Section	What we cover
2.1.	Data warehouse concepts: the four defining properties, OLTP against OLAP, ETL
2.2.	Multidimensional data models: cubes, cuboids, star and snowflake schemas, OLAP operations
2.3.	Data warehouse architecture: the four views, the three tiers, and how one gets built
2.4.	Data warehouse implementation: cube computation, indexing, server architectures
2.5.	Developments in data cube technology
2.6.	From data warehousing to data mining
Lab.	Lab 2 in Google Colab: build a star schema and a cube from the Saigon Bean files

Lecture 2 roadmap and outcome

1

Say what makes a data warehouse different from the transaction system it draws from, and why a firm pays for both

CLO2

2

Model a business as dimensions and measures and draw the star schema that supports it

CLO2

3

Perform roll-up, drill-down, slice, dice and pivot, and say what question each one answers

CLO2

4

Use Python to turn transaction tables into a cub and read the result

CLO3

Where this chapter sits

Steps two and three of the knowledge discovery (KDD) process



File	Rows	In warehouse language	What it contributes
saigon_bean_transactions.csv	97,239	The fact table	The measures: quantity, discount, line_total
saigon_bean_stores.csv	12	A dimension table	store, district, store type, seats
saigon_bean_products.csv	17	A dimension table	product, category, list price
order_time inside the fact table	91 days	A dimension waiting to be built	hour, day, month, quarter



2.1

Data Warehouse Concepts

2.1 Data Warehouse Concepts

What is a data warehouse?

DEFINITION, W. H. INMON

A data warehouse is a subject-oriented, integrated, time-variant and nonvolatile collection of data in support of management decision making.

Subject-oriented

Organized around subjects such as customer, product and sales, rather than around the daily operations of one system

Integrated

Built from many heterogeneous sources, made consistent in naming, encoding and units before it is loaded

Time-variant

Keeps history. Every record carries a time element, and the horizon is years rather than the current value

Nonvolatile

Physically separate from the operational stores. It is loaded and read, never updated transaction by transaction

Four adjectives, and a firm that drops any one of them has built something else. Drop time-variant and you have a reporting database. Drop nonvolatile and you have slowed down your cash registers.

2.1 Data Warehouse Concepts

Subject-oriented and integrated

Subject-oriented

- Organized around major subjects: customer, product, store, sales
- Built for analysis, so it leaves out data that is useless to a decision maker
- Gives a simple, consolidated view of one subject, not a faithful copy of one system

Integrated

- Constructed from relational databases, flat files and transaction records
- Cleaning and integration happen before loading, never after
- Naming, encoding, units and measurement scales are made consistent

INTEGRATION, SEEN IN THE SAIGON BEAN FILES

The channel column arrives with six spellings for three real values: in-store, app, delivery, and also In-Store, App, Delivery. A warehouse resolves that once, at load time, and every report built on it afterwards agrees. Without the warehouse, each analyst resolves it privately, differently, and silently, and two reports on the same quarter stop reconciling.

2.1 Data Warehouse Concepts

Time-variant and nonvolatile

Time-variant

- The time horizon is much longer than that of an operational system
- An operational database holds the current value. A warehouse holds the history
- Every key structure carries a time element, explicitly or implicitly
- Operational keys often do not

WHY HISTORY CHANGES THE QUESTION

The Saigon Bean transaction system can tell you what a store sold yesterday. It cannot tell you whether the campus stores have been drifting toward app orders over three years, because it never kept the answer. Almost every question a manager finds interesting is a question about change.

Nonvolatile

- A physically separate store, transformed from the operational data
- Operational updates do not reach it
- Only two operations matter: the initial load, and the query
- No transaction processing, no recovery, no concurrency control machinery

AND WHY SEPARATION IS NOT LAZINESS

A complex analytical query can scan millions of rows. Run it against the live transaction database and the queue at the counter gets longer. Separation protects the operation from the analysis, and it protects the analysis from the locking that the operation needs.

2.1 Data Warehouse Concepts

OLTP against OLAP

	OLTP (<i>On-line transactional processing</i>)	OLAP (<i>On-line analytical processing</i>)
Users	Clerk, IT professional	Knowledge worker, manager
Function	Day-to-day operations	Decision support
Design	Application-oriented, entity-relationship	Subject-oriented, star or snowflake
Data	Current, guaranteed up to date	Historical, accuracy maintained over time
Usage	Repetitive, known in advance	Ad hoc, written on the spot
Access	Read and write, short transactions	Read mostly, complex queries
Unit of work	One short transaction	One complex query
Records touched	Tens	Millions
Size	100 MB to GB	100 GB to TB and beyond
Metric	Transaction throughput	Query throughput, response time

2.1 Data Warehouse Concepts

Why a separate data warehouse?

The case for separation

- Both systems perform better when each is tuned for its own workload
- The two have very different data: historical and consolidated against current and detailed
- Consolidation needs cleaning and integration that an operational system will not do
- Missing data, and data from outside the firm, has to live somewhere

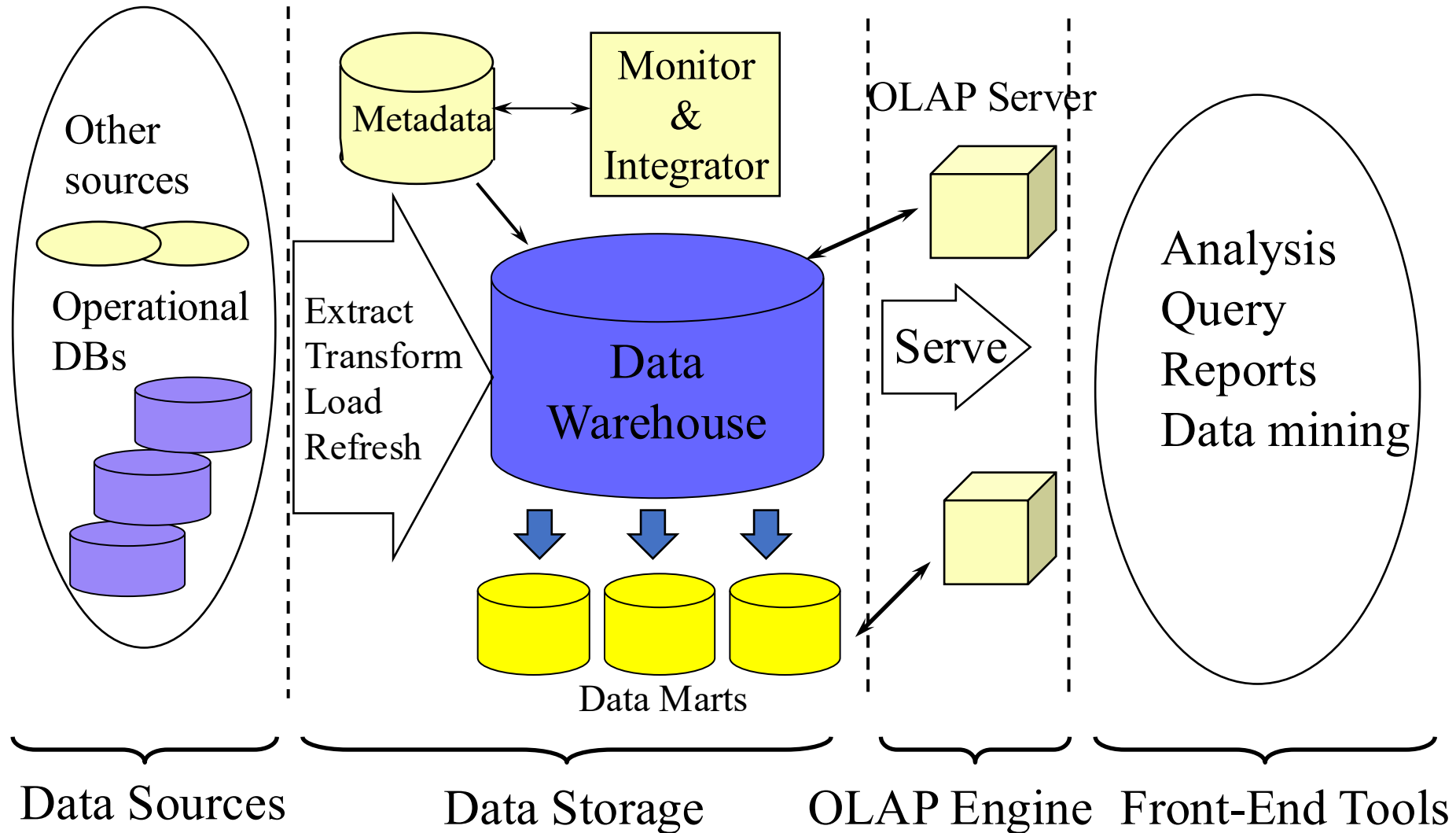
The case against, and it is real

- A second copy of the data costs money and staff
- Every copy can go stale, and reconciling two sources is somebody's job
- Modern engines can run some analytics on operational data directly
- The trend is toward fewer copies, and the arguments above have not gone away

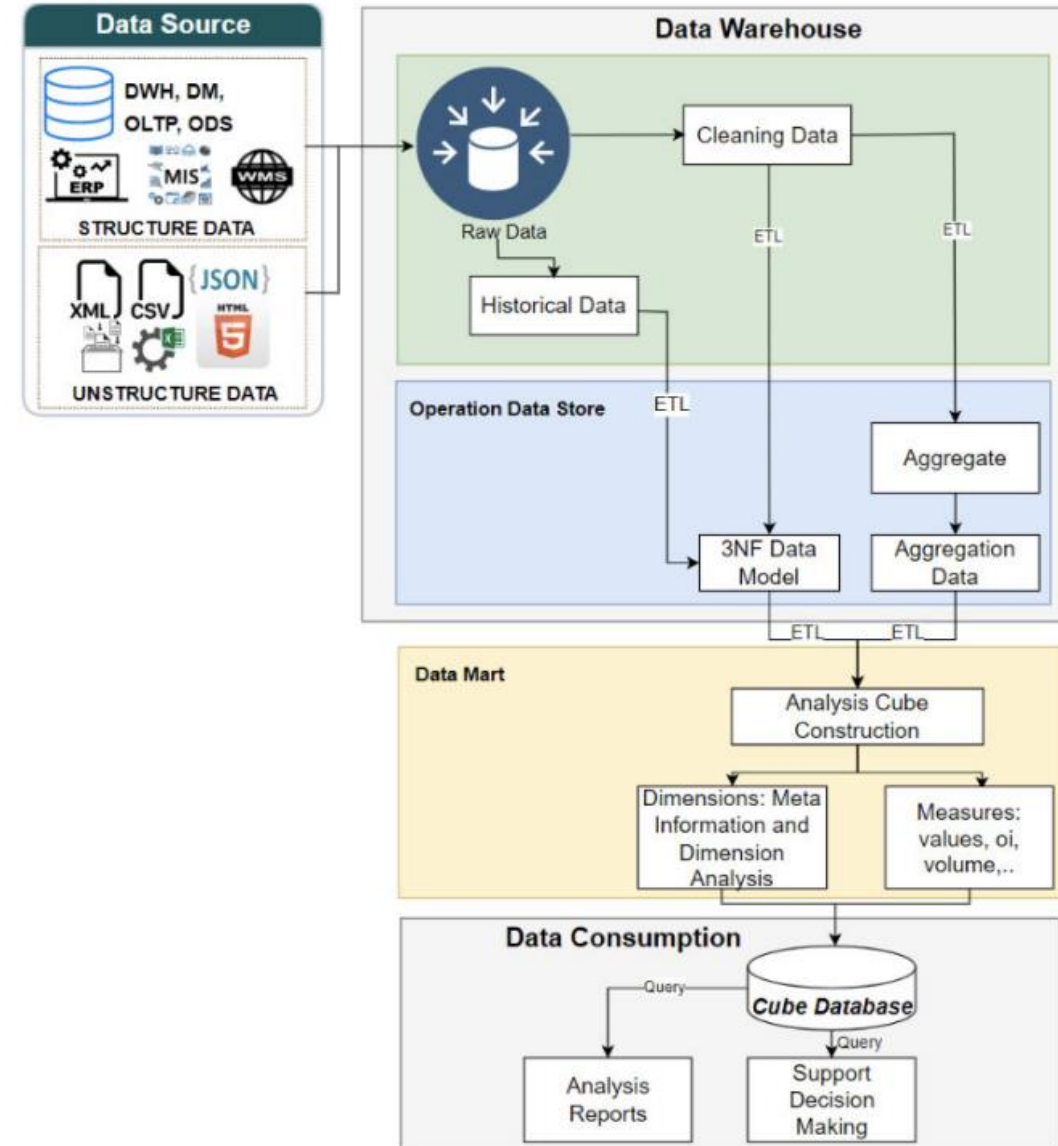
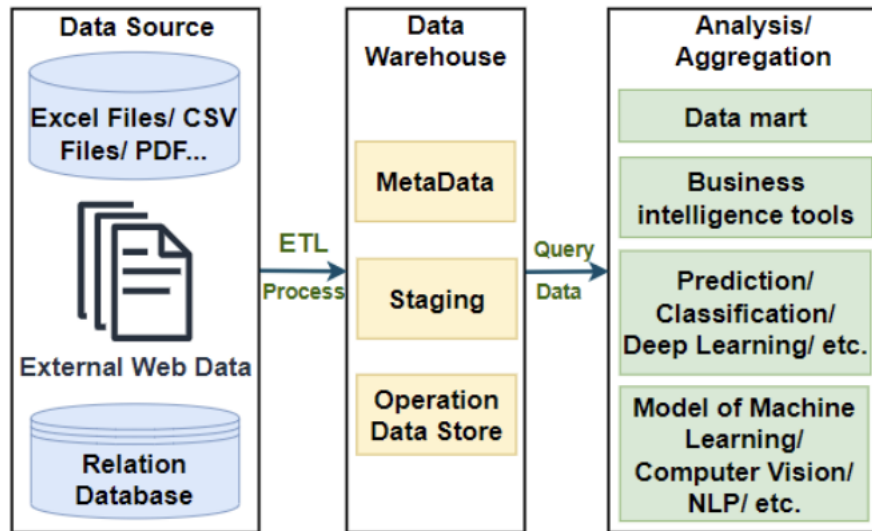
THE DECIDING ARGUMENT

Different functions, different data. A warehouse can carry decision-support data that the operational system does not hold at all: history that was overwritten, sources from outside the firm, and the reconciled version of a field that three systems each spell differently.

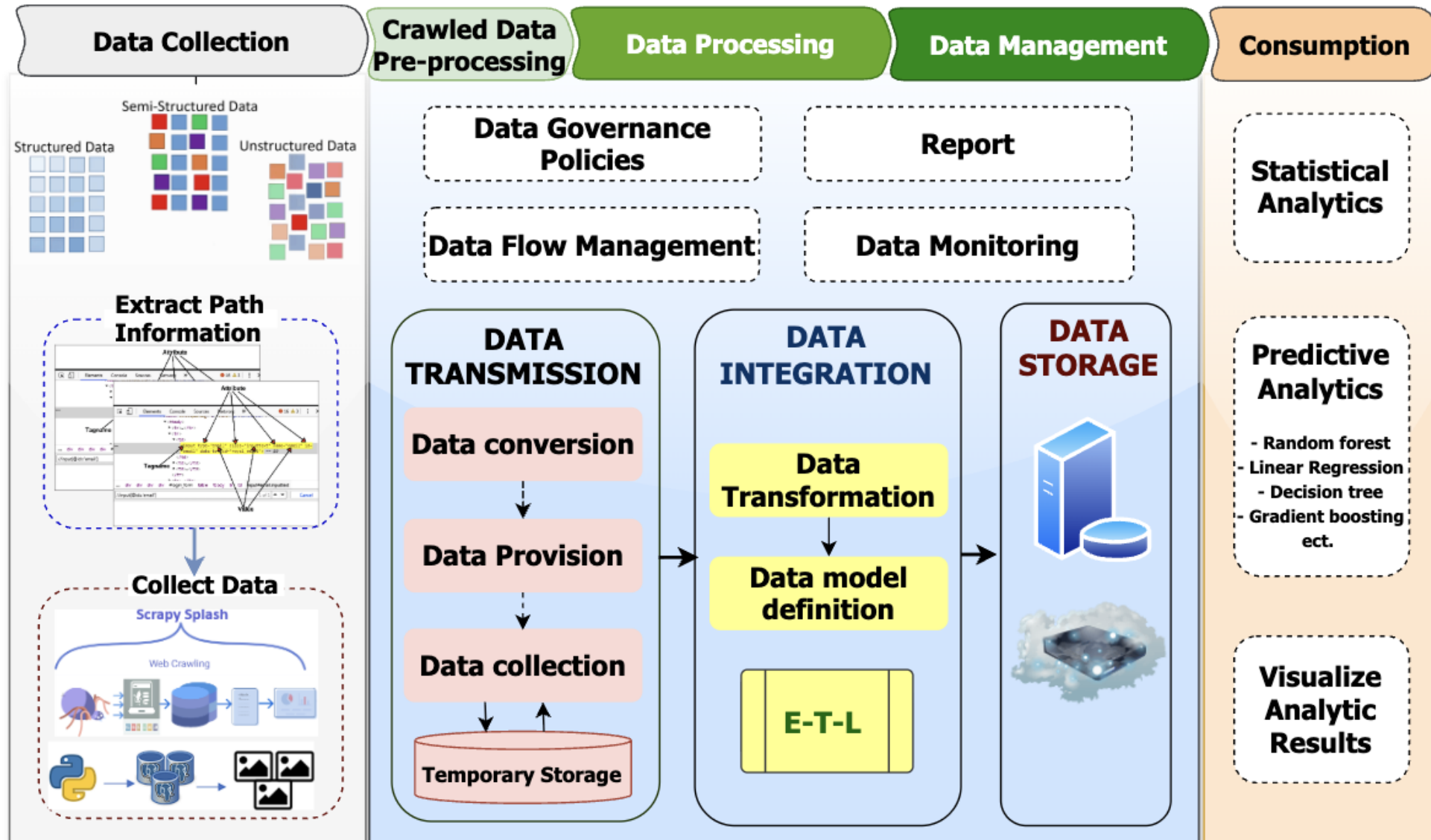
Data Warehouse: A Multi-Tiered Architecture



Data Warehouse: A Multi-Tiered Architecture

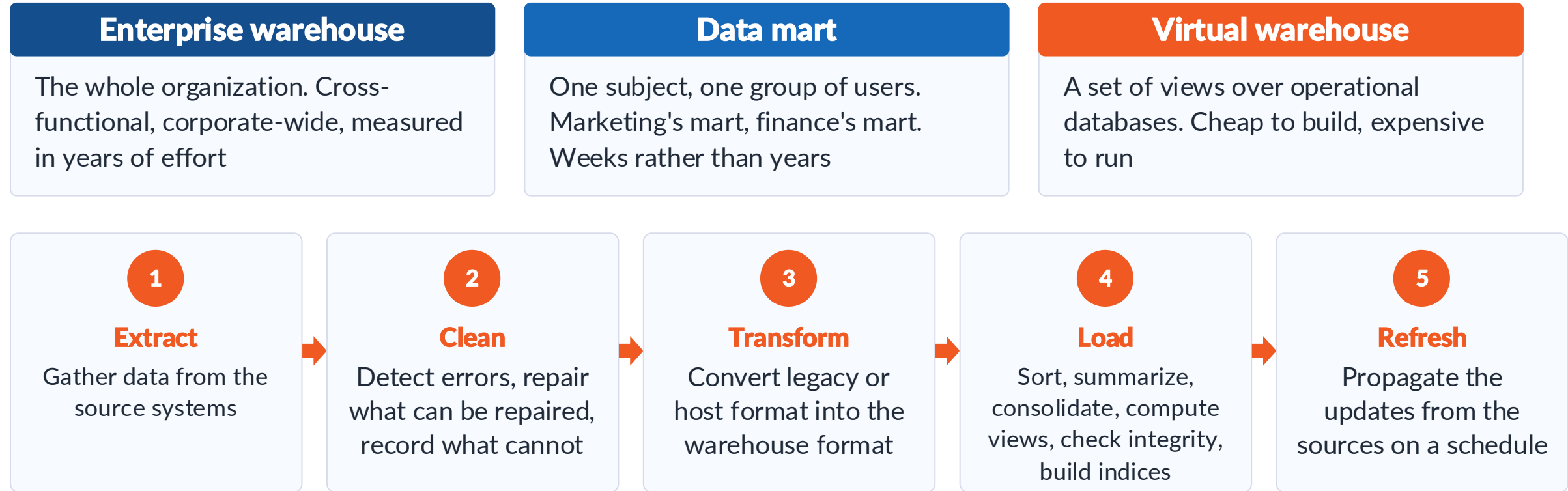


Data Warehouse: A Multi-Tiered Architecture



2.1 Data Warehouse Concepts

Three warehouse models, and how data gets in



2.1 Data Warehouse Concepts

The metadata repository

- **Meta data** is the data defining warehouse objects. It stores:
- Description of the **structure** of the data warehouse
 - schema, view, dimensions, hierarchies, derived data defn, data mart locations and contents
- **Operational** meta-data
 - data lineage (history of migrated data and transformation path), currency of data (active, archived, or purged), monitoring information (warehouse usage statistics, error reports, audit trails)
- The **algorithms** used for summarization
- The **mapping** from operational environment to the data warehouse
- Data related to **system performance**
 - warehouse schema, view and derived data definitions
- **Business data**
 - business terms and definitions, ownership of data, charging policies

Practice 1

- 1 For each statement, decide whether it describes OLTP or OLAP, and name the row of the comparison table that settles it. (a) A cashier voids a line on a receipt. (b) The head of retail compares pastry margin by district across three quarters. (c) The app checks whether a loyalty card is valid. (d) Finance rebuilds last year's revenue after a definition change.
- 2 Saigon Bean runs one transaction database. The marketing team wants to analyse three years of baskets. Write the case for a separate warehouse in four sentences, one per defining property, addressed to a finance director who is asking why one database is not enough.
- 3 A data mart is built for marketing and a second for finance. Six months later they report different revenue for the same quarter. Give two causes that this chapter predicts, and one thing the firm should have built first.



2.2

Multidimensional Data Models

Cubes, cuboids, schemas, and the five OLAP operations

2.2 Multidimensional Data Models

From tables and spreadsheets to data cubes

The model

- A data warehouse is modelled as a multidimensional data cube
- A cube has dimensions, and it has measures
- A dimension is a perspective on the business: time, store, product, channel
- A measure is what you aggregate: quantity, revenue, discount

Definition

- Fact table: the measures, plus the keys of the dimensions
- Dimension table: everything you know about one perspective
- Cell: one combination of dimension values, holding the aggregated measure

“CUBE” DOES NOT MEAN THREE

A cube in this sense has as many dimensions as the business needs. Three is the number you can draw, not the number you build. An n-dimensional cube is often called an n-D base cuboid.

2.2 Multidimensional Data Models

Saigon Bean as a cube

DIMENSIONS	CONCEPT HIERARCHY	IN OUR FILES
time	hour < day < month < quarter < year	91 days, June to August 2026
store	store < district < city	12 stores, 7 districts, 4 store types
product	product < category	17 products, 4 categories
channel	channel	in-store, app, delivery

MEASURES

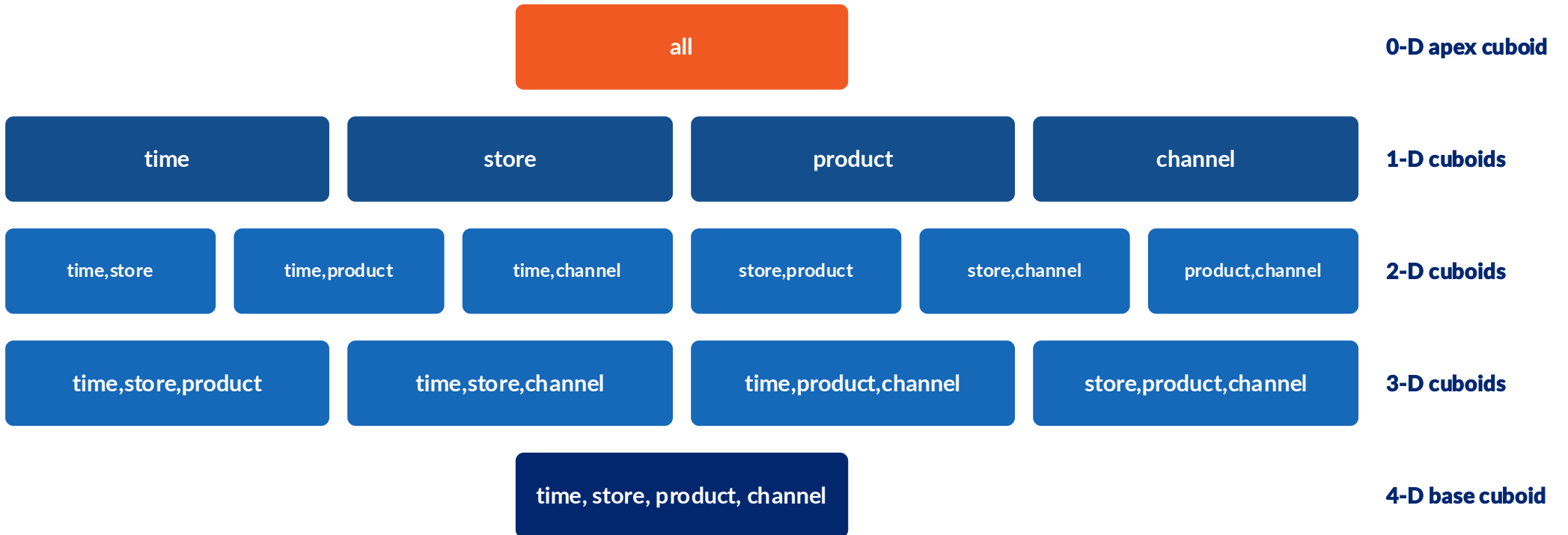
quantity, discount and line_total, all additive over every dimension.
Derived measures follow: revenue, order count, average basket, revenue per seat.

THE DIMENSION NOBODY GAVE YOU

There is no time table in the three files. order_time is a single timestamp, and hour, day, month and quarter have to be derived from it before any roll-up is possible.

2.2 Multidimensional Data Models

A cube is a lattice of cuboids



HOW MANY CUBOIDS

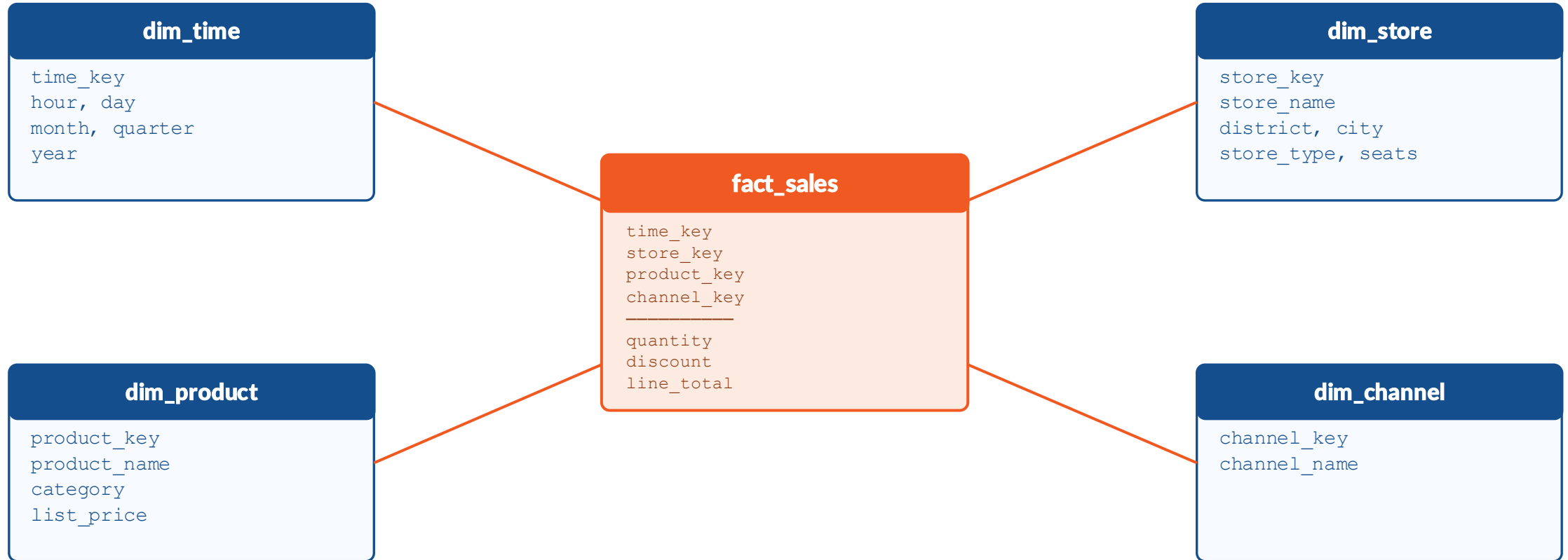
Four dimensions with no hierarchy give $2^4 = 16$ cuboids. Adding the hierarchies gives $\prod (L_i + 1) = 5 \times 4 \times 3 \times 2 = 120$.

WHY THE NUMBER MATTERS

Every cuboid you materialize costs storage and load time

2.2 Multidimensional Data Models

Star schema for Saigon Bean



One fact table in the middle, holding the measures and nothing but foreign keys besides. One table per dimension around it, each denormalized into a single flat table. That shape is a star.

2.2 Multidimensional Data Models

Snowflake schema and fact constellation

Snowflake schema

- The dimension tables are normalized, so each one splits into a hierarchy of smaller tables
- dim_store becomes store, then district, then city, joined by keys
- Saves storage and removes update anomalies
- Costs joins, and therefore query time. Most warehouses accept the redundancy instead

Fact constellation

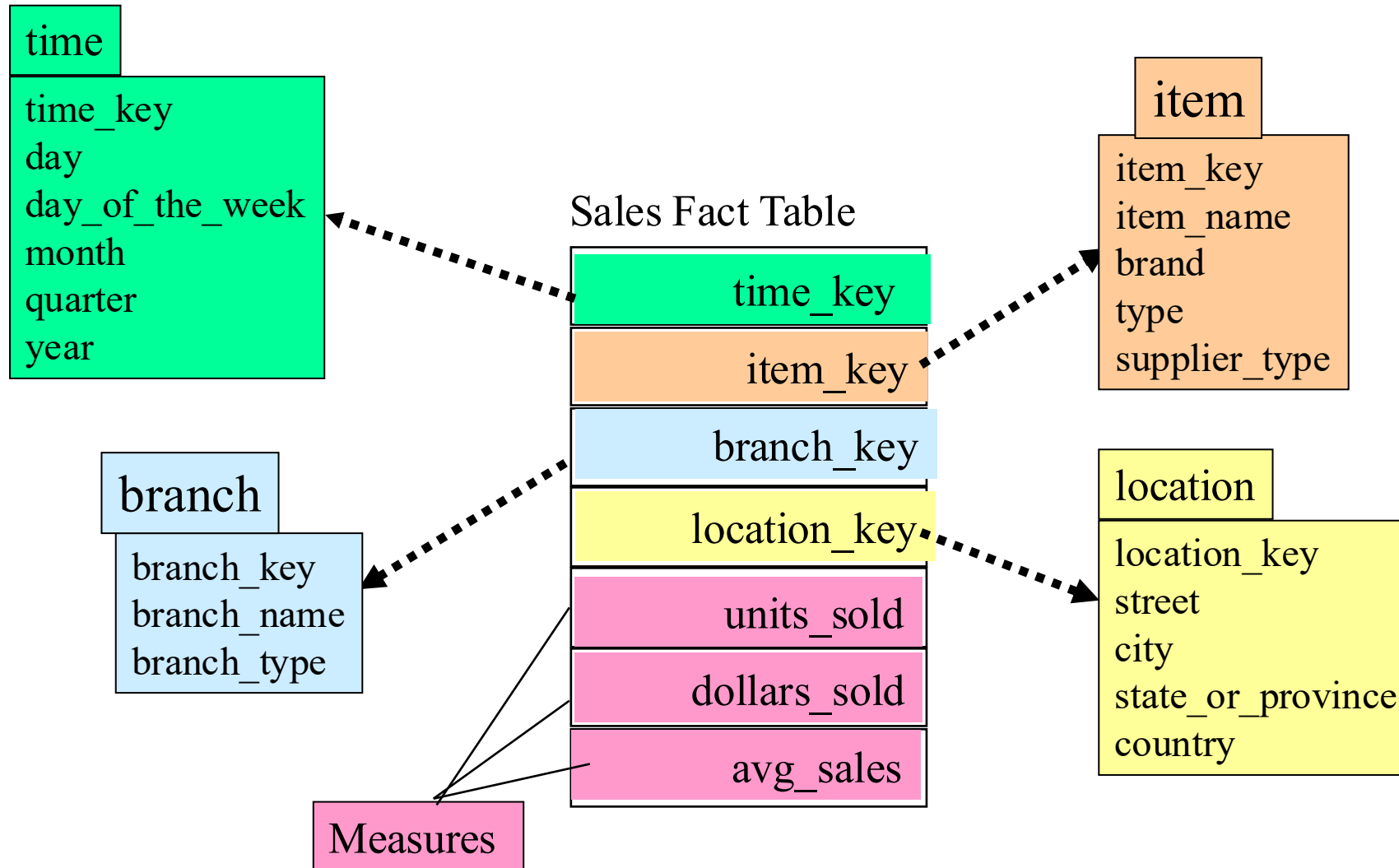
- More than one fact table, sharing dimension tables between them
- Also called a galaxy schema
- Saigon Bean: a sales fact table and a staffing fact table, both using dim_time and dim_store
- The shared dimensions are what make the two comparable

```
STAR      fact_sales — dim_store(store_key, store_name, district, city, store_type, seats)

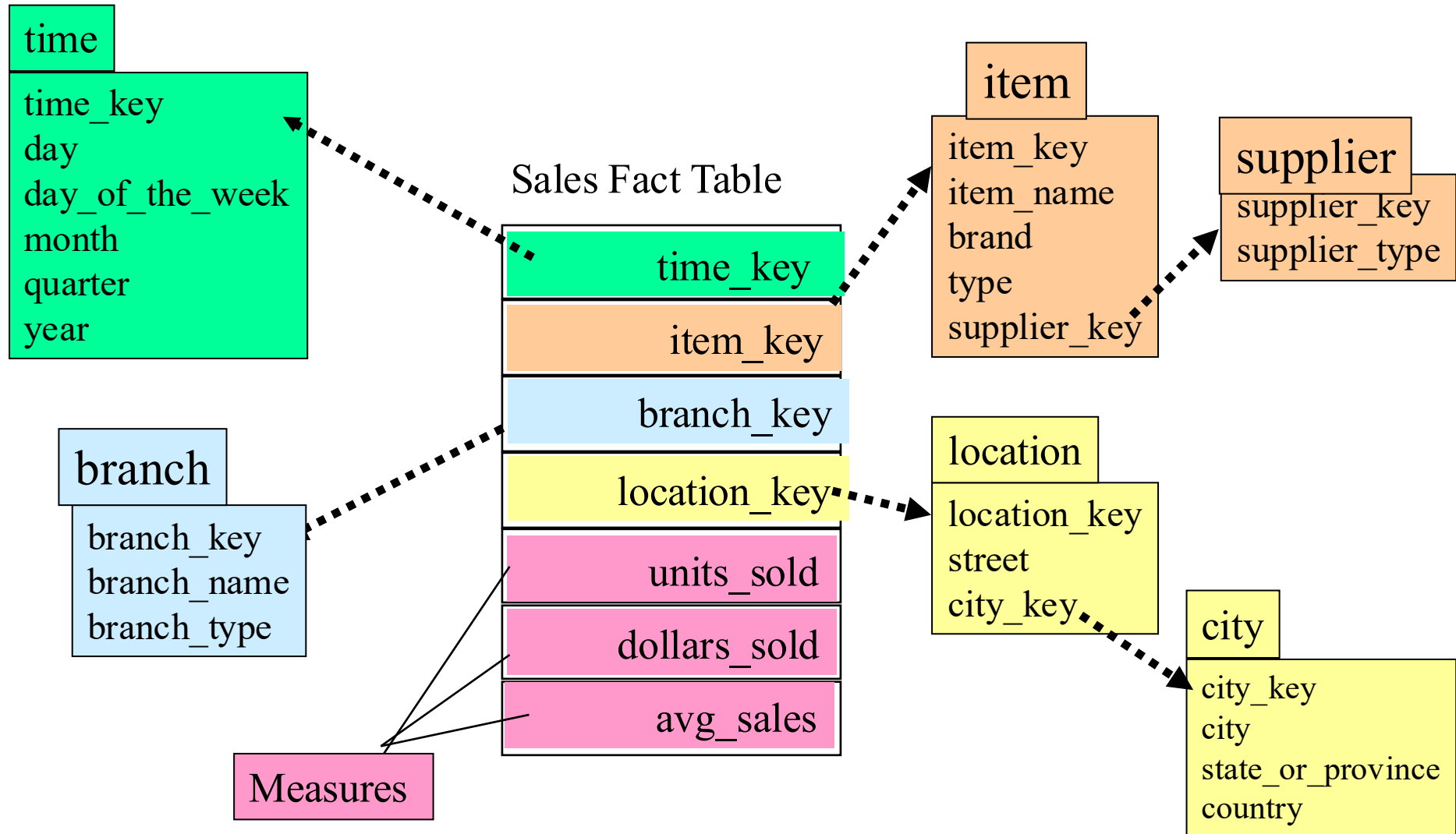
SNOWFLAKE fact_sales — dim_store(store_key, store_name, district_key, store_type, seats)
                        └─ dim_district(district_key, district_name, city_key)
                           └─ dim_city(city_key, city_name)
```

Choose the star unless storage is the binding constraint. Analysts write fewer joins, queries run faster, and the redundancy is cheap on a dimension table with twelve rows.

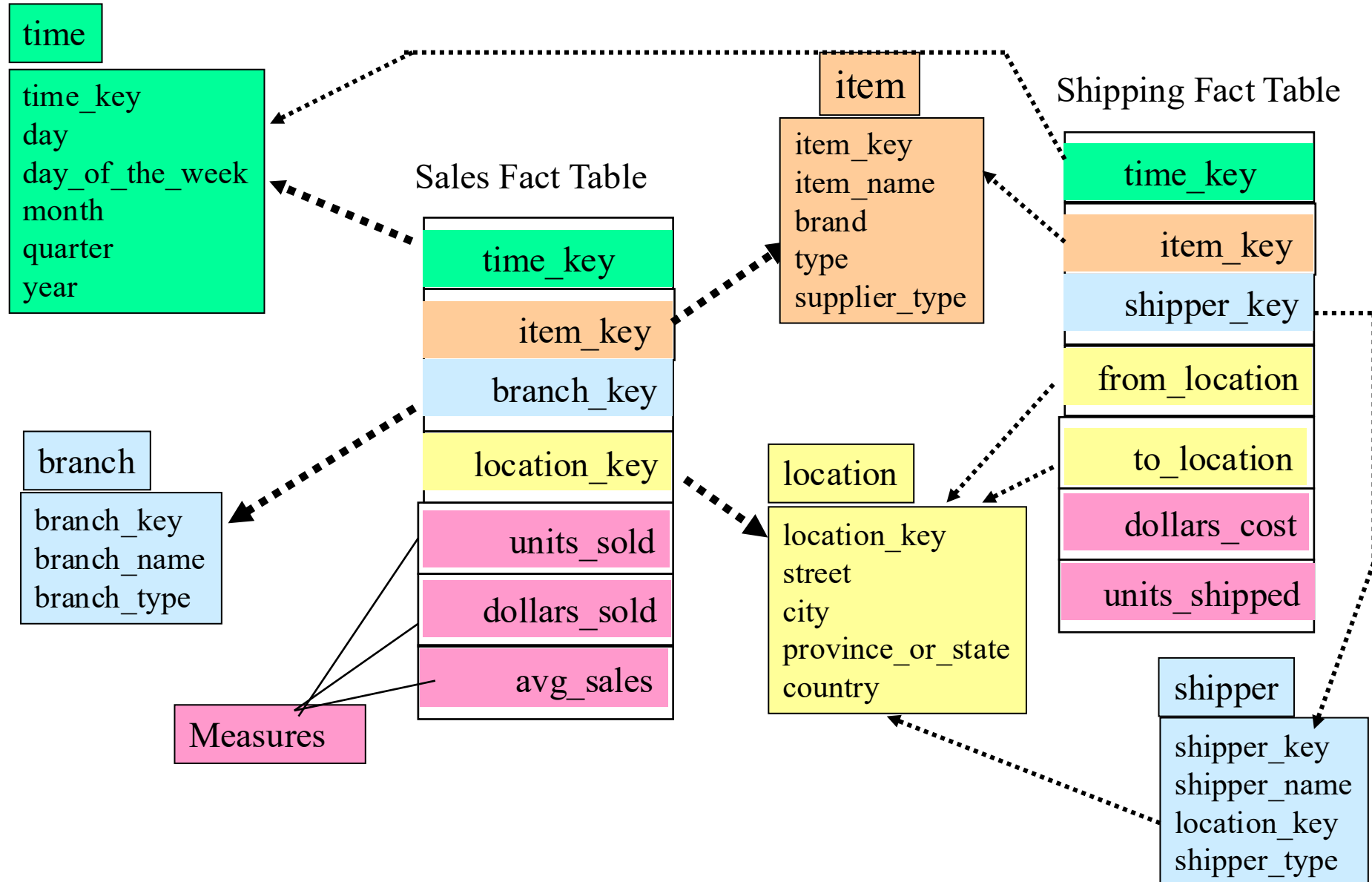
Example of Star Schema



Example of Snowflake Schema



Example of Fact Constellation

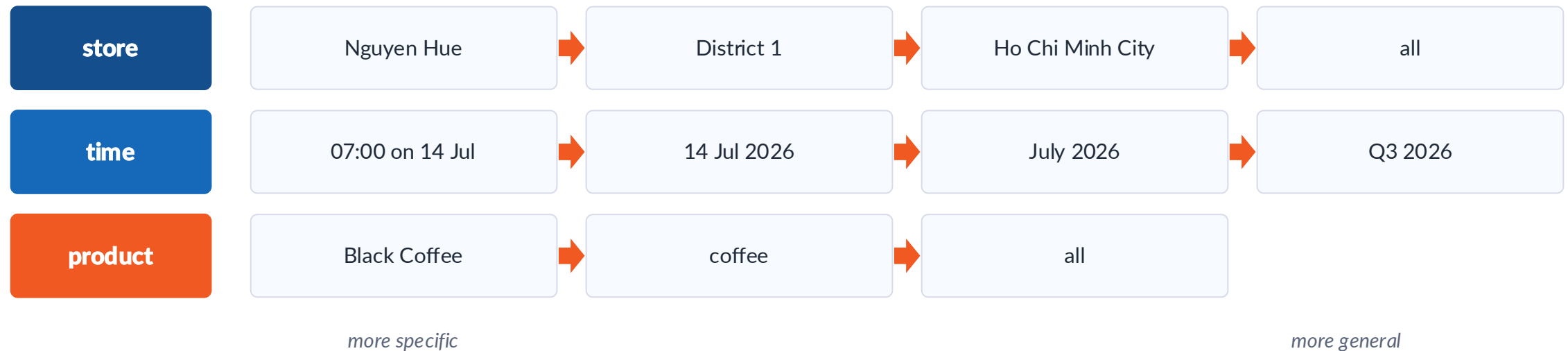


2.2 Multidimensional Data Models

Concept hierarchies

DEFINITION

A concept hierarchy defines a sequence of mappings from a set of low-level concepts to more general, higher-level concepts. It is what makes roll-up and drill-down possible.



Hierarchies can be given by a schema, given by hand for a particular attribute, or generated automatically from the data. Every one of them encodes a business decision about what counts as the same thing.

2.2 Multidimensional Data Models

Three categories of measure

Distributive

The function can be computed by partitioning the data, computing each part, and combining the results.

`count, sum, min, max`

Algebraic

Computable from a fixed number of distributive measures.

`avg = sum / count,
standard deviation`

Holistic

No constant bound exists on the storage needed to describe a sub-aggregate.

`median, mode,
rank, distinct count`

THE SAME CELL, THE THREE KINDS: mall × coffee × June

Distributive: 10,442 order lines, summing to 640,348,950 dong, the largest of them 118,000. Algebraic: the average line is $640,348,950 / 10,442 = 61,324$ dong, and both inputs were already in the cell. Holistic: the median line is 52,000 dong, and computing it needed every one of the 10,442 values, not a summary of them.

A warehouse can precompute distributive and algebraic measures cheaply. A holistic measure usually has to be recomputed from the base data, which is why dashboards quote averages and rarely quote medians.

2.2 Multidimensional Data Models

A real data cube: revenue in million VND

June 2026	coffee	tea	pastry	merchandise	all products
mall	640.3	193.3	310.6	14.2	1,158.4
transit	181.1	61.3	54.5	1.4	298.3
residential	107.5	32.2	48.3	3.3	191.3
campus	84.1	22.9	31.9	2.2	141.1
all store types	1,013.0	309.7	445.3	21.1	1,789.1

WHAT YOU ARE LOOKING AT

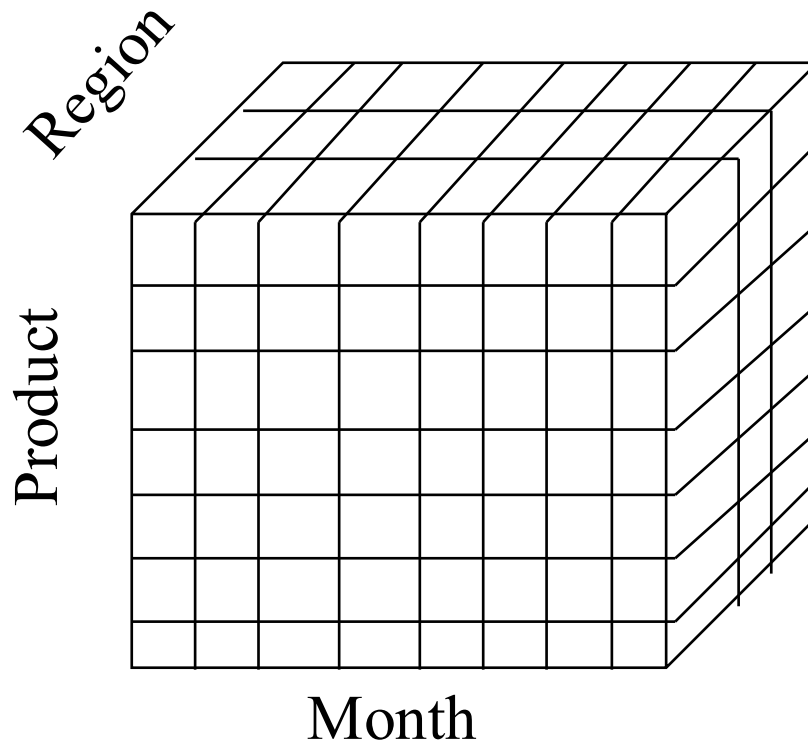
One slice of a three-dimensional cube: store_type × category × month. Four by four by three is 48 base cells, and June is one of the three slices. The margins are not decoration: each one is a cell of a lower-dimensional cuboid.

THE NUMBER IN THE CORNER

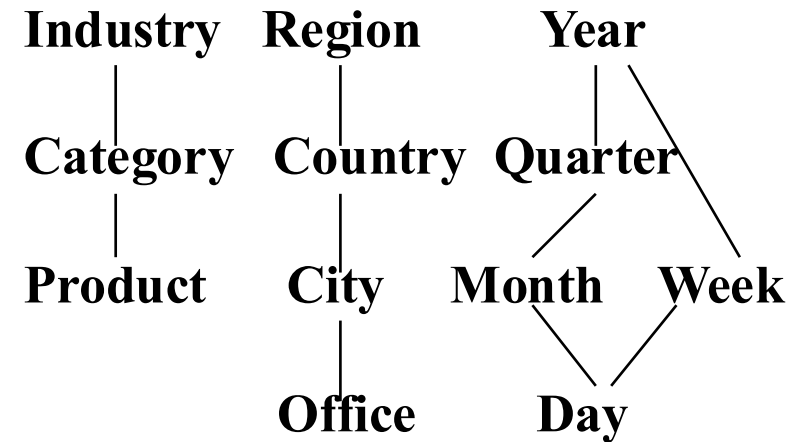
1,789.1 million dong is the apex of this slice. Roll June, July and August together and you reach 5,563 million, which is the apex cuboid of the whole cube: one cell, one number, the entire chain.

2.2 Multidimensional Data Models

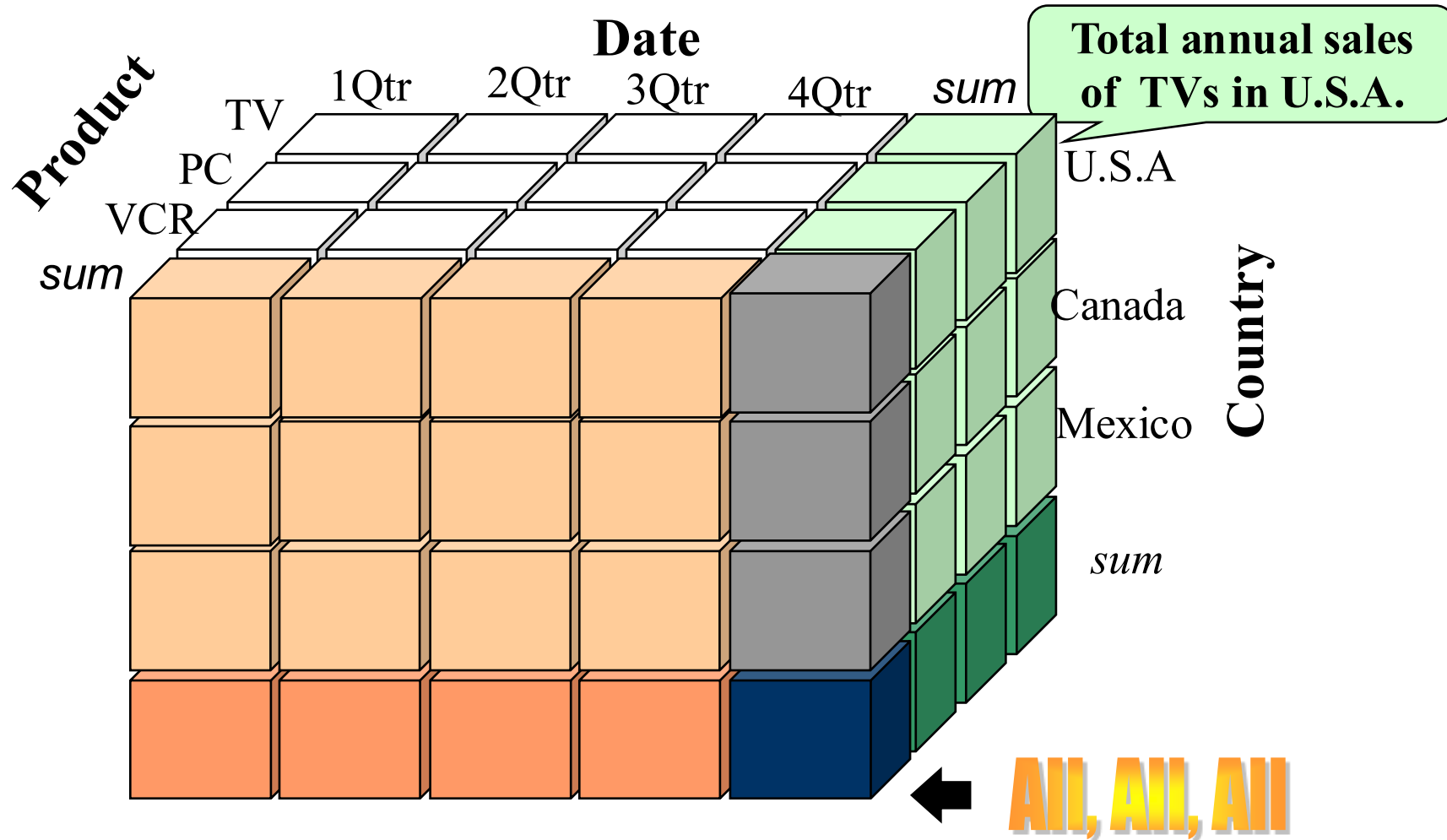
- Sales volume as a function of product, month, and region



Dimensions: *Product, Location, Time*
Hierarchical summarization paths



A Sample Data Cube



2.2 Multidimensional Data Models

The five OLAP operations

Roll-up

Climb a hierarchy or drop a dimension. Aggregate upward.

```
store → district →  
city  
District 1 = 2,015.7  
whole city = 5,563.1
```

Drill-down

The reverse: descend a hierarchy or add a dimension.

```
quarter → month  
Jun 1,789.2  Jul  
1,864.1  
Aug 1,909.9
```

Slice

Fix one dimension to a single value. The result has one dimension fewer.

```
category = coffee  
leaves store_type ×  
month,  
a 4 × 3 table
```

Dice

Fix a subrange on two or more dimensions. The result is a smaller cube.

```
store_type ∈ {mall,  
transit}  
category ∈ {coffee,  
tea}  
month ∈ {Jul, Aug}
```

Pivot

Rotate the axes. The data does not change, the presentation does.

```
swap rows and columns,  
or turn a 3-D cube  
into  
a series of 2-D planes
```

Five operations, and a manager using a dashboard performs all five without knowing any of the names.

Practice 2

- 1 Using the June table on slide 22: (a) which OLAP operation produced the figure 1,013.0? (b) which produced 1,789.1? (c) write the operation that would take you from the mall row to the six individual mall stores. Name the operation and the hierarchy level in each answer.
- 2 Saigon Bean adds a staff roster with hours worked per store per day. Draw the resulting schema. Is it a star, a snowflake or a fact constellation? Name every table that the two fact tables share, and say what question the sharing makes answerable.
- 3 The chain wants median basket value by store by month on its dashboard, refreshed hourly. Explain, using the three categories of measure, why this request is more expensive than average basket value, and offer one alternative that costs less.

SUBMISSION

One Markdown cell per exercise in your Lab 2 notebook.



2.3

Data Warehouse Architecture

2.3 Data Warehouse Architecture

Four views of a warehouse design

Top-down view

Chooses the information the warehouse should hold at all. Driven by what the business has to decide, not by what the source systems happen to store.

Data source view

Exposes what each operational system captures, at what grain, in what format. Often documented in the ER model of that system.

Data warehouse view

The fact tables and dimension tables as they will exist: the star schema, the grain of the fact table, the pre-computed aggregates.

Business query view

What the end user sees from the analyst's desk. If this view is awkward, the warehouse will not be used, however correct it is.

Four views, four groups of people, and four sets of assumptions that have to be reconciled before anyone writes a line of SQL.

2.3 Data Warehouse Architecture

The three-tier architecture

Tier 3

Front-end tools

Query and reporting, analysis, data mining. This is where the analyst sits.

Power BI, Tableau, notebooks

Tier 2

OLAP server

Presents the multidimensional view and answers cube queries. ROLAP, MOLAP or HOLAP.

the cube engine

Tier 1

Warehouse database server

The relational store holding the warehouse and the data marts, fed by extract, clean, transform, load and refresh.

the warehouse itself, plus
metadata

READ IT FROM THE BOTTOM

Data flows up: source systems feed tier 1, tier 1 feeds tier 2, tier 2 answers tier 3. Requirements flow down, and they usually arrive from tier 3 first.

2.3 Data Warehouse Architecture

How a warehouse gets designed and built

Two ways to approach the design

- Top-down: start from overall design and planning. Good when the technology is mature and the business problem is well understood
- Bottom-up: start with experiments and prototypes. Good early, when neither of those holds
- In practice: a planned top-down architecture, delivered bottom-up in increments

THE RECOMMENDED APPROACH

Define a high-level enterprise model first, in weeks rather than months. Then build independent data marts inside it, in parallel, and distribute them. The enterprise model is the contract that keeps the marts consistent.

The four design steps

- Choose a business process to model: sales, orders, shipments
- Choose the grain of the fact table: what one row means
- Choose the dimensions that apply to each fact row
- Choose the measures that will populate each fact row

GRAIN IS THE DECISION THAT BINDS

In our files one fact row is one line of one receipt. Choose the order instead and you can never answer a product question. Choose the day and you lose the morning peak. Grain cannot be refined later without reloading everything.

2.3 Data Warehouse Architecture

Three kinds of warehouse usage

Information processing

- Querying, basic statistics, reporting
- Tables, charts, graphs
- Answers a question that somebody already knew how to ask

Analytical processing

- Multidimensional analysis of warehouse data
- Roll-up, drill-down, slice, dice, pivot
- The analyst supplies the hypothesis and the cube verifies it

Data mining

- Knowledge discovery from hidden patterns
- Associations, classification, clustering, prediction
- The pattern was not specified in advance

THE SAME LINE AS LECTURE 1

Query, report, or mine. Chapter 1 drew this boundary with a coffee-chain example and located it at who supplies the hypothesis. The warehouse serves all three, and only the third is data mining. A firm that buys a cube and expects discovery has bought the wrong thing.

2.4

Data Warehouse Implementation

2.4 Data Warehouse Implementation

Efficient cube computation

```
compute cube sales_cube as
  select store_type, category, month, sum(line_total)
  from   fact_sales
  cube by store_type, category, month

→ computes all  $2^3 = 8$  cuboids at once
```

The operator, and its price

- One statement builds every cuboid in the lattice
- Cost grows exponentially in the number of dimensions
- This is the curse of dimensionality in its warehouse form

THE ARITHMETIC THAT DECIDES THE DESIGN

Total cuboids = $\prod (L_i + 1)$, where L_i is the number of levels in dimension i . Saigon Bean with four dimensions and the hierarchies given in section 2.2: $5 \times 4 \times 3 \times 2 = 120$ cuboids. Add one dimension with three levels and it becomes 480. Add a customer dimension with ten thousand members and the base cuboid alone stops fitting anywhere convenient.

No materialization

Compute every cuboid on demand. Cheap to store, slow to query.

Full materialization

Precompute all of them. Fast to query, and the storage explodes.

Partial materialization

Precompute a chosen subset. The only workable answer at scale.

2.4 Data Warehouse Implementation

Indexing OLAP data: the bitmap index

How it works

- One bit vector per distinct value of the indexed attribute
- The n-th bit is 1 when row n carries that value
- A query becomes a bitwise AND or OR, which hardware does extremely fast

WHEN IT IS THE RIGHT INDEX

Best where the attribute has few distinct values, which is exactly what a dimension attribute looks like: four store types, four categories, three channels, twelve stores. Low cardinality means few bit vectors and heavy compression.

store_type	mall	campus	resid	transit	
row 1	mall	1	0	0	0
row 2	transit	0	0	0	1
row 3	campus	0	1	0	0
row 4	mall	1	0	0	0

WHEN IT IS THE WRONG ONE

High cardinality destroys it. A bitmap index on `customer_id` with 4,777 members needs 4,777 bit vectors, almost all of them nearly empty. That is what B-tree indices are for.

2.4 Data Warehouse Implementation

Join indices, and the three OLAP server architectures

The join index

- Registers the joinable rows of two relations in advance
- In a star schema, it links each dimension value to the fact rows that carry it
- The join is then a lookup rather than a scan, and star joins are the dominant cost in warehouse queries

WHY IT PAYS HERE AND NOT IN OLTP

A join index has to be maintained whenever the data changes. In an operational system that cost lands on every transaction. In a warehouse the data changes only at load time, so the index is built once per refresh and read all day.

ROLAP

Relational OLAP. The cube lives as tables in a relational store, with optimisation for star queries.

Scales to the largest data. Slower on complex aggregations.

MOLAP

Multidimensional OLAP. A dedicated array-based engine holds the cube directly.

Very fast on dense cubes. Struggles when the cube is sparse.

HOLAP

Hybrid. Detail stays relational, summaries live in the multidimensional engine.

The usual commercial answer, and the usual compromise.



2.5

Developments in Data Cube Technology

2.5 Developments in Data Cube Technology

Computing only the cells that matter

The iceberg cube

- Materialize only the cells whose measure clears a minimum threshold
- Most cells of a large sparse cube hold a count of zero or one
- The threshold is the iceberg condition, and the surviving cells are the tip
- Typical saving is one to two orders of magnitude

```
compute cube sales_iceberg as
  select store, product, month,
         count(*), sum(line_total)
  from   fact_sales
  cube by store, product, month
  having count(*) >= 50
```

Shell fragments

- Precompute cuboids over small groups of dimensions
- Assemble the answer at query time from the fragments

Closed cubes

- Store one representative for every group of cells sharing a measure
- Lossless, and far smaller than the full cube

Discovery-driven exploration

- The cube pre-computes exception indicators
- The interface leads the analyst toward cells that deviate

Discovery-driven exploration is the closest a cube gets to mining: it still cannot form a hypothesis, and it can point at where one is needed.

2.5 Developments in Data Cube Technology

Where the technology went after the textbook

Columnar storage

Store a table by column instead of by row. Aggregating one column no longer reads the others, and compression works far better on a single-type column. This is why a modern engine can scan a billion rows without a precomputed cube.

Cloud warehouses

Storage separated from compute, so the two scale independently and a firm pays for a query rather than for a machine. Snowflake, BigQuery and Redshift are the names students will meet.

The lakehouse

One store holding raw and modelled data together, queried directly. It attacks the copy that section 2.1 argued for, and the argument about reconciliation has not been settled by it.

What did not change

Dimensions, measures, grain, hierarchies and the five OLAP operations. The storage engine changed. The model on this slide deck did not.

Learn the model rather than the product. Vendors will keep changing, and a firm will still ask you what one row of the fact table means.



2.6

From Data Warehousing to Data Mining

2.6 From Data Warehousing to Data Mining

From OLAP to OLAM

Why mining benefits from a warehouse

- The data is already integrated, cleaned and consistent, which is most of the work
- The information processing infrastructure already exists: access, integration, consolidation, transformation, storage, web access
- OLAP-based exploratory analysis lets the analyst steer the mining, rather than run it blind
- On-line selection of the mining function: switch task without rebuilding the data

On-line analytical mining

- OLAM couples mining to the cube, so mining runs at any level of abstraction
- Roll up, mine, drill down, mine again
- The analyst changes the granularity of the question without changing the pipeline
- Mining is integrated into the tool the business already uses

THE SAIGON BEAN VERSION

Cluster the twelve stores at the store level and you get twelve objects and a weak result. Roll the cube up to store type and the same algorithm has four well-populated groups. Drill down inside the mall group to district and the question changes again. The cube is what makes that movement cheap, and the level of abstraction is a parameter of the mining, not a fixed property of the data.

A warehouse is not required for data mining, and it removes most of the reasons a mining project fails.

Practice 3

- 1 Saigon Bean's dashboard runs slowly. Three candidate fixes are proposed: materialize every cuboid, add a bitmap index on `customer_id`, and move from ROLAP to MOLAP. For each, say whether it would help, and why. Exactly one of the three is a clearly bad idea.
- 2 The chain adds a customer dimension with 4,777 members and a promotion dimension with 40 campaigns, each with two levels. Recompute the number of cuboids, and give one sentence on what that number means for the refresh window.
- 3 Write, in plain English, one question about the Saigon Bean business that a cube can answer in a second, and one that no cube can answer however it is built. Explain what makes the second question different.

SUBMISSION

Practice 1 to Practice 3 together form Task 5 of Lab 2. Exercise 3 is the one the final exam tends to reuse.

Quick check

1. Which property explains why a warehouse can answer a question about last year?

- A. Subject-oriented
- B. Integrated
- C. Time-variant
- D. Nonvolatile

2. Going from revenue by store to revenue by district is which operation?

- A. Drill-down
- B. Roll-up
- C. Slice
- D. Pivot

3. Which measure cannot be computed by combining sub-aggregates?

- A. sum
- B. count
- C. average
- D. median

4. A bitmap index on customer_id, 4,777 distinct values, would be

- A. Ideal, because the column is a key
- B. A poor fit, because cardinality is too high
- C. Required for any star join
- D. Equivalent to a join index

Answers: 1C, 2B, 3D, 4B. Question 3 is the one to discuss, because the average looks holistic and is not.

Lab 2 in Google Colab and submission

Task	What to produce
Task 1	Build the time dimension from order_time: hour, day, month, quarter. Then assemble a star schema from the three files: one fact table and four dimension tables
Task 2	Integrate before you load: resolve the six channel spellings into three, and state your rule for the rows that cannot be repaired
Task 3	Build the store_type × category × month cube and reproduce the June table from slide 22 exactly
Task 4	Perform all five OLAP operations on your cube, one cell per operation, and write one sentence per result saying what a manager would do with it
Task 5	Practice 1 to Practice 3, one Markdown cell per exercise

SUBMISSION

File named StudentID_FullName_DM_Lab02.ipynb on the LMS, plus the Colab share link set to view. Runtime then Run all before you submit, so the notebook reproduces from a clean state.

Summary

- 1 A data warehouse is subject-oriented, integrated, time-variant and nonvolatile, and it exists to support decisions rather than to run the business.
- 2 OLTP and OLAP differ in users, design, data, access pattern and the metric each is tuned for. A firm runs both because no single design serves both well.
- 3 The multidimensional model has dimensions and measures. A cube is a lattice of cuboids, from the apex holding one number to the base cuboid holding the finest grain.
- 4 The star schema puts one fact table at the center and one flat table per dimension around it. The snowflake normalizes the dimensions, the constellation shares them between fact tables.
- 5 Measures are distributive, algebraic or holistic, and the category decides whether the measure can be precomputed or must be rebuilt from base data.
- 6 Roll-up, drill-down, slice, dice and pivot are the five operations, and a concept hierarchy is what makes the first two possible.
- 7 Implementation is a trade between storage and query time: partial materialization, iceberg cubes, bitmap and join indices, ROLAP against MOLAP against HOLAP.
- 8 The warehouse is the precondition for mining, not a substitute for it. A cube verifies a hypothesis the analyst supplied, and Chapter 3 begins the work of finding one.



ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC KINH TẾ - LUẬT

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
UNIVERSITY OF ECONOMICS AND LAW

THANK YOU!

Next: Chapter 3, Characterization and Comparison

